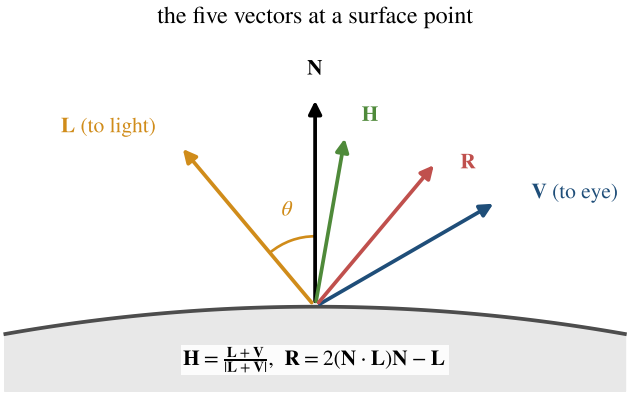
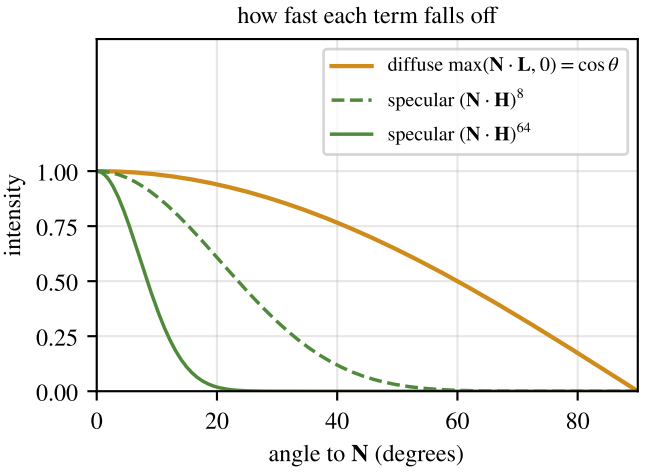

the five vectors at a surface point
N
L (to light)
H
R
V (to eye)
θ
H = (L+V)/|L+V|,  R = 2(N · L)N − L


how fast each term falls off
diffuse max(N · L, 0) = cos θ
specular (N · H)^8
specular (N · H)^64
intensity
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
angle to N (degrees)